
The Shadow Anthology: Reconstructing Rejected Tokens as Text, and Why Their Differences Are Only Probability Depth

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Abstract

Every token a sampled language model emits is a choice: the sampler draws one token from a distribution and discards the record of what it declined. We present a method that, from a single generation call, recovers this discarded record and renders it as a readable text, the *shadow poem*, composed of the runner-up token at every position the model actually wrote. The shadow shares the original’s exact length and lineation, costs no additional forward passes, and turns one generation into a paired reading of the poem that was written and the poem that was rejected at each decision point. A shadow poem is a counterfactual comb rather than a sample: each shadow token is conditioned on the *written* prefix, so it is the pointwise record of roads not taken, not a text the model would have produced end to end. On a corpus of 128 poem/shadow pairs per condition (SmolLM2-1.7B-Instruct; four temperatures, shadow ranks 1–3, and a gated condition), we report three findings. First, a measurement: the sampler’s editorial footprint is small and strongly temperature-dependent; at $T=0.3$, 57% of positions have no usable second candidate at all, so across most of the text there is nothing for the sampler to author. Second, a controlled negative result: every lexical difference between a poem and its nearest-rejected sibling that reaches significance is explained by probability depth rather than by the sampler’s choice, established with a rank-ladder control and family-wide correction over all 88 tests; no imagery or tone metric survives at rank 1 in any temperature condition. Third, an argument about agency and attribution: the information needed to credit the sampler exists at generation time in every deployed system and is discarded before display, and for a growing class of systems (providers that withhold log-probabilities, and reasoning models that compose text inside a hidden chain of thought and transcribe it) the sampler’s authorship cannot be read at all. All results are reproducible from committed artifacts with no model or GPU.

1 Introduction

When a language model writes a poem under stochastic decoding, the text the reader receives is not the only text that was in contention: at every position, the sampler held a distribution over next tokens, drew one, and threw the rest away. The draw is an editorial act, selecting among candidates the model itself proposed, yet it is uncredited; no deployed interface shows the reader what was declined, and the record needed to reconstruct it is discarded before display.

This paper makes that act visible, and asks where creative agency sits inside a generation. From a single call, we log the sampler’s own post-temperature, post-truncation distribution at each step and reconstruct the *shadow poem*: the text formed by taking, at every position, the highest-probability token the sampler did *not* choose. Because substitution happens position-by-position against the

written text, the shadow inherits the original’s token count, line breaks, and rhythm exactly, and the two texts form a naturally paired unit with the same prompt, seed, model, and length. The construction generalizes: rank- k shadows take the k -th rejected candidate, a *gated* shadow swaps only the most contested positions, and forced-prefix continuation from contested points yields a branching *anthology* of genuinely resampled near-poems off a single seed.

We are explicit about the central object. A shadow poem is a **counterfactual comb**, not a sample: each shadow token is conditioned on the prefix the model actually wrote, so it records the road not taken at each point rather than a coherent alternative trajectory (the branching anthology is the trajectory-level construction). This distinction governs which claims the instrument can support.

The instrument was built to test a hypothesis: that the chosen poem and its nearest-rejected sibling differ systematically in imagery, tone, or lexical risk; that the sampler, in aggregate, is an editor with a taste. **That hypothesis failed**, and we report the failure as a result. What survives is more interesting:

1. **A method** (§3): from one generation call, recover the counterfactual record the sampler destroys, as a readable text with the original’s exact structure. Free, exact, seedable, no extra forward passes.
2. **A measurement** (§5.1): how much of a poem is available to be decided is itself a function of temperature. At $T=0.3$, most tokens were never chosen over anything.
3. **A controlled negative result** (§5.2): poem–shadow lexical differences track probability depth, not choice, shown by a rank-ladder control and family-wide correction over all 88 tests.
4. **An agency argument** (§5.4): which systems can be traced at all is itself a finding; closed APIs without log-probabilities and reasoning models that transcribe verse composed inside a hidden chain of thought both place the sampler’s authorship beyond reading.

2 Related work

Truncation-based stochastic decoding, including top- k [Fan et al., 2018] and nucleus sampling [Holtzman et al., 2020], is standard for open-ended generation, and a large literature studies how decoding strategy shapes text quality and diversity. That literature compares corpora produced under *different* strategies; we instead compare a single generation against the alternatives the *same* draw rejected, holding everything else fixed. Per-token log-probabilities are widely used for calibration, hallucination detection, and watermarking, but to our knowledge the runner-up token stream has not previously been rendered as a text object in its own right and subjected to paired literary measurement against the chosen text. Where machine-poetry work typically evaluates generated poems against human poems or each other, our paired design asks a narrower question: conditioned on everything the model brought to the page, what difference did the draw make? Lexical metrics are backed by published psycholinguistic norms [Brysbaert et al., 2014, Warriner et al., 2013]; multiple-comparison control uses Holm [Holm, 1979] and Benjamini–Hochberg [Benjamini and Hochberg, 1995].

3 Method

3.1 Tracing the sampler

We generate with a hand-rolled sampling loop rather than a convenience API, because we need the distribution the sampler *actually drew from* at each step, after temperature scaling and top- p /top- k truncation, which standard generation utilities discard. Each trace records, per position, the chosen token, its sampling log-probability, and the top candidates of the truncated, tempered distribution (up to 20). Structural and template tokens are excluded from candidate lists: a shadow poem must never contain a literal `<|im_end|>` mid-verse, and such tokens were never poetic alternatives. From a trace we derive, per position: the distribution’s entropy (bits), the top-1/top-2 margin (nats), whether the chosen token was the argmax (*off-argmax* positions are exactly where the sampler overrode the model’s preference), and whether a *usable alternative* existed, a distinct, non-structural rejected candidate available for substitution. Positions with the top two candidates within 0.1 nats are flagged as *coin flips*: places the generation came nearest to going otherwise.

3.2 Shadow, gated shadow, rank ladder

The **rank- k shadow** substitutes, at every position with a usable alternative, the k -th-ranked rejected candidate; positions without one keep the written token, as do line breaks and mid-word subword pieces in the rendered reading (the statistics substitute at those too). The rank-1 shadow is the nearest-rejected sibling. The **gated shadow** substitutes only at the n most contested positions (smallest probability gap), leaving the rest of the poem intact; with $n=12$ this swaps roughly 10% of tokens and remains a fluent poem. The shadow’s *divergence rate*, the fraction of positions actually substituted, doubles as a corpus-level statistic: the share of the text at which any decision existed.

3.3 Branching anthologies

Where the backend supports forced-prefix continuation, we can go beyond the comb: at each of the most contested positions, force the rejected candidate and let the model continue freely, yielding a tree of fluent near-poems that genuinely fork at the sampler’s closest calls. Branching is presented as method and used to construct paired fluent texts for the designed (not yet run) human study; **no corpus-level branching results are claimed**, as no branch corpus is committed.

3.4 Measurement and statistics

Each text is scored on eleven lexical metrics: concreteness, abstract ratio, sensory density, imagery, valence, arousal (norm-backed), and rarity, repetition, lexical risk, type–token ratio, and mean word length; metrics with insufficient lexicon coverage report as missing rather than neutral. Every comparison is paired, since poem and shadow share prompt, seed, model, and length, so we use a two-sided sign-flip permutation test on within-pair differences (exact in its null, assumption-free), with bootstrap confidence intervals and Cohen’s d_z effect sizes. Within each arm the eleven tests are Holm-corrected; across the study all tests are pooled into one family and corrected again (§5.2). Holm was pre-specified as primary; Benjamini–Hochberg is secondary.

4 Experimental setup

All committed results use **SmolLM2-1.7B-Instruct** [Allal et al., 2025] run locally, with 16 prompts $\times$ 8 seeded samples = **128 poem/shadow pairs per arm**. Arms: four temperatures ($T \in \{0.3, 0.7, 1.0, 1.3\}$, top- $p=0.95$), shadow ranks 1–3 (re-analyses of the identical $T=1.0$ traces, preserving the pairing; regenerating per rank would compare different poems and destroy the design), and one gated condition (top-12 contested positions, same traces). Published norms are used throughout, and every table and figure below regenerates from the committed outputs with no model, no GPU, and no network.

One caveat is inherited by all cross-arm comparisons: `max_tokens` was 224 for $T \leq 1.0$ but 400 for the $T=1.3$ arm, where 79% of generations hit the cap; mean lengths run 91 to 350 tokens across arms and several metrics are length-sensitive. Within-arm poem/shadow pairing is unaffected (both sides of every pair have identical token counts), but cross-arm effect-size comparisons are not clean, and we do not make them.

5 Results

5.1 The sampler’s editorial footprint

Table 1 and Figure 1 give the study’s positive result: **how much of the poem is available to decide is itself a function of temperature**. At $T=0.3$, only 11.9% of tokens are off-argmax and 57% of positions have no usable second candidate whatsoever; those tokens were never chosen over alternatives. The sampler’s authorship does more than weaken at low temperature: across most of the text, there is nothing for it to author. At $T=1.3$ the picture inverts (93.5% of positions carry a real alternative, most tokens are non-argmax draws), but the poems there are long, cap-truncated, and formally degraded, so this is not a free lunch of agency.

Table 1: Sampler footprint by temperature (128 poems per arm). *Off-argmax*: positions where the sampler did not take the model’s preferred token; under greedy decoding this is 0 by construction, which is what makes it the sampler’s footprint rather than the model’s. *Usable-alt.*: positions with a distinct, non-structural rejected candidate. *Coin-flip*: top two candidates within 0.1 nats. The $T=1.3$ arm ran at `max_tokens=400`.

T	mean tokens	entropy (bits)	margin (nats)	off-argmax	usable-alt.	coin-flip	hit cap
0.3	91.1	0.415	1.493	11.9%	43.1%	2.94%	0.8%
0.7	117.0	1.259	1.329	28.9%	72.1%	2.29%	5.5%
1.0	151.0	1.960	1.184	41.0%	82.6%	1.87%	27.3%
1.3	349.5	2.658	1.070	53.9%	93.5%	5.14%	78.9%

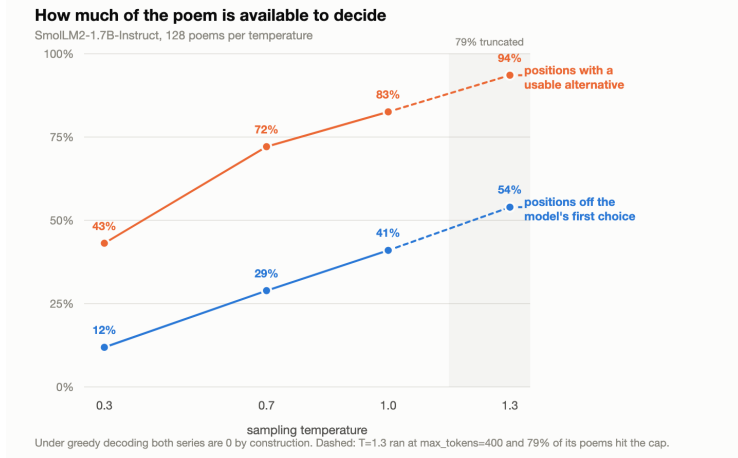


Figure 1: **The sampler’s editorial footprint scales with temperature.** Share of off-argmax positions and of positions with a usable alternative, across four temperatures; 128 poems each. At $T=0.3$, 57% of positions have no usable second candidate. The $T=1.3$ arm is dashed: 79% of its generations hit the token cap.

5.2 The poem and its shadow do not differ in imagery or tone

The original hypothesis predicted systematic imagery, tone, or risk differences between the poem and its rank-1 shadow. Two controls show that every significant lexical difference is instead an artifact of probability rank.

Rank-ladder control. The shadow is *defined* as a lower-probability token, and token probability correlates with corpus frequency, repetition, and word length, so some poem/shadow differences are guaranteed by the selection rule alone. To separate selection artifact from editorial signal, we compare the poem against rank-1, rank-2, and rank-3 shadows built from the identical traces (Figure 2). If a metric’s delta grows monotonically with rank, it tracks how far down the ranking one looked, not what the sampler chose. Seven of eleven metrics do exactly that (rarity, repetition, risk, TTR, concreteness, abstract ratio, imagery); type-token ratio and mean word length additionally *reverse sign* between ranks 1 and 3 (TTR $+0.032 \rightarrow -0.060$; word length $+0.123 \rightarrow -0.129$), which no account of a consistent editorial preference can accommodate. Sensory density, valence, and arousal are null at every rank.

Family-wide correction. A study with eight arms and eleven metrics runs ~ 88 tests, and correcting within each arm while reading starred rows across arms is a well-known way to manufacture findings (~ 4 expected false positives at $\alpha=0.05$ even under the global null). Pooling all tests into one family (88 tests; 77 unique after collapsing the rank-1 arm, which is the $T=1.0$ analysis repeated verbatim), **26 survive Holm and 31 survive Benjamini–Hochberg**, and every survivor is a depth-artifact form metric (rarity, repetition, risk, TTR, word length, concreteness, abstract ratio) or *imagery* at rank 3, already identified as the depth artifact by the rank ladder. **No imagery or tone metric survives at rank 1 in any temperature condition.** Valence, the only imagery/tone metric to approach

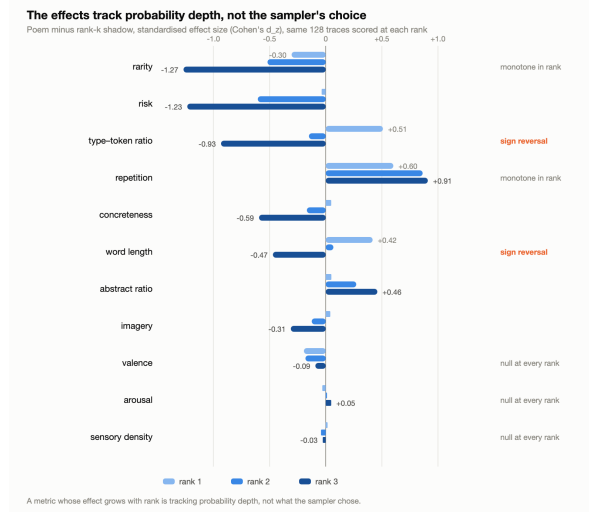


Figure 2: **The poem/shadow differences track probability depth, not choice.** Effect size (Cohen’s d_z), poem minus rank- k shadow, same 128 traces at every rank. Seven of eleven metrics grow monotonically with rank; TTR and word length reverse sign. A metric whose effect grows with rank measures probability depth, not what the sampler chose.

significance in any single arm ($T=1.3$, within-arm Holm $p=0.084$), does not survive family-wide correction (Holm $p=0.50$); arousal reaches significance in no arm at all. Five rows that are significant within their own arm fail family-wide, and these are exactly the rows a within-arm-only analysis would have published (Figure 3).

The negative result, stated plainly: *the sampler’s choice does not measurably change what kind of poem you get. What it changes is whether there was a poem to choose between at all* (§5.1).

5.3 The method, made readable

A single trace supports three readings: the poem as written; the gated shadow, which swaps only the sampler’s closest calls ($n=12$, roughly 10% of tokens) and remains a fluent poem while surfacing exactly where the generation nearly went otherwise; and the full shadow, which reads as a double rather than a poem, since each substitution is conditioned on the written prefix rather than the shadow’s own; that is the object, not a failure of it. The released reading interface renders any trace this way, with every rejected alternative, all drawn from the sampler’s support at that position, visible on hover.

5.4 Which models can be traced

The method’s preconditions are themselves informative, because they fail in two distinct ways in deployed systems. **Withheld records:** some provider APIs expose no log-probabilities or top- k alternatives, so the runner-up token is unrecoverable from any response, and rejected assistant prefix closes the black-box workaround of forcing a prefix and resampling to estimate the distribution. **Relocated choices:** a subtler failure arises with reasoning models, which compose the poem *inside their chain of thought* and then transcribe it, so the visible poem span is a near-deterministic copy. In an exploratory probe of 18 hosted models (probe log not committed, so reported as exploratory), the poem span showed mean entropy of ≈ 0.06 bits with 0% off-argmax positions, against ≈ 1.0 bits and 28.5% in the reasoning span: the sampler made no choices in the poem span, and reconstructing its “rejected” tokens would report alternatives never in contention. Our pipeline aborts on such spans (entropy < 0.15 bits) rather than producing a confident, meaningless result.

Both failures sharpen the agency argument. **The information needed to credit the sampler exists at generation time in every deployed system and is discarded before display.** Where a provider withholds it, attribution is impossible by policy; where an architecture moves the real choosing into

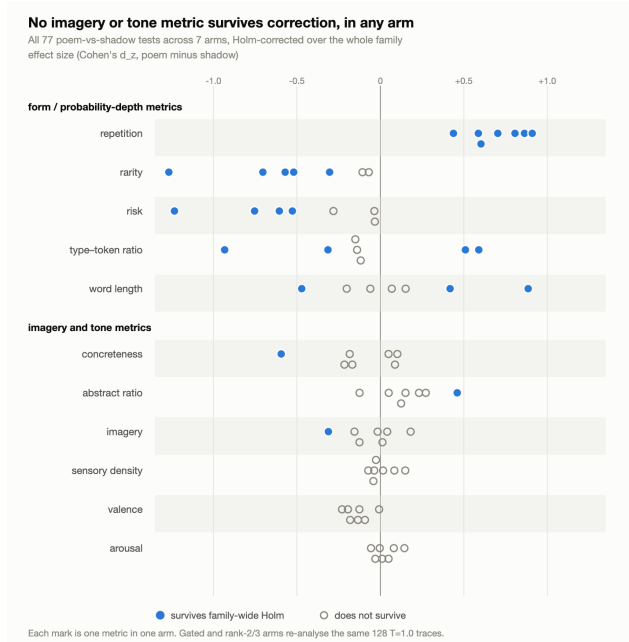


Figure 3: **No imagery or tone metric survives family-wide correction.** All 77 unique poem-vs-shadow tests across seven arms; filled marks survive family-wide Holm, hollow do not. Every survivor is a form metric or *imagery* at rank 3, the depth artifact of Figure 2. Within-arm correction alone would have published five further claims.

a hidden span, impossible by construction. Either way, an editorial act occurs in every sampled generation, and the reader has no access to it.

6 Discussion and conclusion

The framing that motivated this work, the sampler as an uncredited editor executing a choice at each token, survives the negative result in revised form. The sampler is not an editor with a *taste*: conditioned on the written prefix, its choices do not push the poem toward measurably different imagery, tone, or risk than its nearest-rejected alternatives. What it governs, jointly with temperature, is the *existence* of choice: at low temperature the poem is largely a transcription of the model’s argmax path; at high temperature nearly every token is a genuine draw. The interesting agency question is therefore not “what kind of editor is the sampler?” but “how much of this text was ever up for decision?”, which the shadow trace answers exactly, per token, from one generation. Neither fact was measurable until the rejected tokens were made into a text. We release the tracer, the shadow and anthology constructions, the reading interface, and every committed run, so that any traceable model’s generations can be read the same way: the poem that was written, beside the poem rejected at every step.

7 Limitations

The shadow is a comb, not a sample: shadow tokens are conditioned on the written prefix, so the shadow records rejected choices, not a text the model would have produced. All committed results are one model at 128 pairs per arm, a case study rather than a survey; the $T=1.3$ arm’s larger token cap contaminates cross-arm effect-size comparisons, which we avoid. A blinded rater study is fully built but has not been run, so reader-imperceptibility is *not* claimed and is stated only as designed future work. Aesthetic quality is not measured: nothing here says either poem is better. No branching corpus is committed; the hosted-model probe of \$5.4 is exploratory. Finally, word-norm metrics cannot see syntax, metaphor, or line-level effects; a null here is a null on lexical signal, not on all literary difference.

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